Computational and Replication Appendix

to "Corrective Overoptimism"

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R.A Computational Methods and Verification

R.A.1 County investment algorithm

At each date t and inherited state $\Omega_t = (A_{At}, A_{Mt}, T_t, \mathbf{z}_t, \overline{\mathbf{z}}_t, t)$, the county infrastructure game is solved on the statutory box $\mathcal{X}_t = \prod_{j=1}^{J}[0, \bar{x}_{jt}] \subseteq \prod_{j=1}^{J}[0, o_{jt}]$. A candidate portfolio $\mathbf{x} \in \mathcal{X}_t$ is admissible if its conditional dated equilibrium (Definition 1) yields positive wages and prices, satisfies Stone–Geary subsistence $w(\ell) + b_t(\omega) \geq \gamma p_A(\ell, \omega)$ across all sites and states, and clears goods, labor, county service, and dividend ledgers with construction demand $I_t(\ell; \mathbf{x})$ absorbed from manufacturing output.

For any trial portfolio $\mathbf{x}$, the inner solver computes the pre-discovery plan—residence $n(\ell)$, sector $e_i(\ell) \in \{0, 1\}$, research probability $\phi_i(\ell) \in [0, 1)$, monetary innovation bill $\Psi_i(\phi_i(\ell); w(\ell), \ell)$, and hired labor $(L_i(\ell), L_T(\ell))$—and then solves the statewise post-discovery allocation $(Y_i, p_i, H_i, \tau_j, b_t)$ across $N_\omega = 32$ stratified spatial discovery draws $\omega \in \{1, \ldots, N_\omega\}$ from the Ornstein–Uhlenbeck Gaussian copula ($d_0 = 0.10$) and Pareto step-size law. Current county return $R_{jt}(\mathbf{x}; \Omega_t)$ equals expected county land rent minus the expected spot cost of committed construction goods. Continuation values $\mathbb{E}_t^\theta[V_{j,t+1}^\theta(\Omega_{t+1})]$ in (57) are represented on the reduced county-level state $(\overline{\mathbf{T}}_{t+1}, \overline{\ln \mathbf{A}}_{A,t+1}, \overline{\ln \mathbf{A}}_{M,t+1}, \mathbf{z}_{t+1}, \overline{\mathbf{z}}_t)$ via a second-order Chebyshev–Smolyak polynomial basis $H_{jt}^{(n-1)}$ iterated backward from the post-1881 balanced-growth continuation value (Appendix A.7.2), where $\overline{T}_{j,t+1} = \overline{T}_{jt} + d_{jt}x_{jt}$ updates county j's average route stock (54) and $\overline{\ln A}_{ij,t+1}$ tracks county-mean log knowledge under (40). Conditional on each draw ω, expectations over the diagnostic law (12) of next-period productivity gaps, $\mathbf{z}_{t+1} \sim \mathrm{TN}_{\mathcal{B}}(\mathbf{m}_t^\theta, \boldsymbol{\Sigma})$ on $\mathcal{B} = \prod_{s\in\{A,M,T\}}[-c\nu_s, c\nu_s]$ with $\mathbf{m}_t^\theta = \boldsymbol{\rho}\mathbf{z}_t + \theta\boldsymbol{\rho}[\mathbf{z}_t - M(\boldsymbol{\rho}\mathbf{z}_{t-1})]$, are evaluated by a $5 \times 5 \times 5$ Gauss–Hermite quadrature rule (125 nodes) mapped to $\mathcal{B}$ via the Cholesky factor of $\boldsymbol{\Sigma}$ and renormalized on the truncated support.

Fixed points of the county best-response map solve the projected pseudogradient system

$$\mathbf{r}_{t,\alpha}(\mathbf{x}) = \mathbf{x} - \mathrm{proj}_{\mathcal{X}_t}\left[\mathbf{x} + \alpha\widehat{\mathbf{g}}_t^\theta(\mathbf{x})\right] = 0, \qquad \alpha > 0, \tag{R.A.1}$$

where $\widehat{\mathbf{g}}_{jt}^\theta(\mathbf{x}) = \partial_{x_j}\widehat{\Pi}_{jt}^\theta(x_j, \mathbf{x}_{-j})$ is county j's unilateral payoff derivative along the implicit conditional equilibrium and differentiated Chebyshev–Smolyak continuation surface. Zeros of $\mathbf{r}_{t,\alpha}(\mathbf{x})$ satisfy the KKT conditions (A.1), equivalently (62), and are unique under Proposition 7.

Numerically, the solver initializes from inherited reference portfolios and alternates damped Anderson-accelerated pseudogradient steps with coordinate-wise unilateral sweeps, evaluating every candidate by recomputing the full conditional dated equilibrium and continuation value.

At a converged candidate $\widehat{\mathbf{x}}$, each county's maximum unilateral deviation gain is checked on $\{0, 1/4, 1/2, 3/4, 1\}\bar{x}_{jt}$ plus local refinement around $\widehat{x}_j$:

$$\epsilon_j(\widehat{\mathbf{x}}) = \max_{u \in [0, \bar{x}_{jt}]} \{\widehat{\Pi}^{\theta}_{jt}(u, \widehat{\mathbf{x}}_{-j}) - \widehat{\Pi}^{\theta}_{jt}(\widehat{\mathbf{x}})\} \leq 10^{-10},$$

before advancing the state to $t+1$ via (54).

R.A.2 Joint-equilibrium differentiation

To differentiate simulated moment paths with respect to the transformed parameter vector z, we differentiate the conditional private equilibrium and the continuous county investment KKT system jointly along each simulated history $b \in \{1, \ldots, B\}$ and date t. Let s_{bt} denote the inherited state, u_{bt} the conditional private-equilibrium unknowns, and $\mathbf{x}_{bt}$ the county investment vector, and let $\mathcal{R}_{bt}$ stack the pre-discovery plan conditions and the finite-volume spot-market relations of Section R.A.3.1:

$$\mathcal{R}_{bt}(u_{bt}, \mathbf{x}_{bt}, s_{bt}, z) = 0, \qquad \overline{\mathbf{g}}^{\theta}_{bt}(\mathbf{x}_{bt}; u_{bt}, s_{bt}, z) \text{ satisfies (A.1).} \tag{R.A.2}$$

Partition the county coordinates into lower-bound ($\mathcal{L}_{bt}$), interior ($\mathcal{I}_{bt}$), and upper-bound ($\mathcal{U}_{bt}$) active sets. Substituting $x_{j,bt} = 0$ on $\mathcal{L}_{bt}$ and $x_{j,bt} = \bar{x}_{j,bt}$ on $\mathcal{U}_{bt}$, the interior equilibrium system is

$$\mathcal{H}_{bt}(q_{bt}, s_{bt}, z) = \begin{bmatrix} \mathcal{R}_{bt} \\ \overline{\mathbf{g}}^{\theta}_{bt,\mathcal{I}_{bt}} \end{bmatrix} = 0, \qquad q_{bt} = (u'_{bt}, \mathbf{x}^{I\prime}_{bt,\mathcal{I}_{bt}})'. \tag{R.A.3}$$

PROPOSITION R.A.1: *Fix a finite set of shock histories and a parameter z_0. At every reached history and date, suppose: (i) $\mathcal{H}_{bt}$ is continuously differentiable with nonsingular Jacobian $\mathcal{H}_{q,bt}$; (ii) bound coordinates satisfy strict complementarity; (iii) the own-concavity and rival-sensitivity bounds in (63) and (64) hold with positive slack; and (iv) all active private-equilibrium faces and state transitions $\mathcal{G}_{bt}$ are locally regular. Then a neighborhood of z_0 preserves the KKT active sets and unique equilibrium branch, and $(u_{bt}, \mathbf{x}_{bt})$ is continuously differentiable on that neighborhood.*

Proof. By the implicit-function theorem applied to (R.A.3), q_{bt} is locally C^1 with fixed active sets under strict complementarity and contraction slack. Induction over the finite horizon and intersection across the B histories yield a common neighborhood of z_0. □

If $c_q(z)$ enumerates the signed complementarity, curvature, and branch slacks with $m_q = c_q(z_0) > 0$ and Lipschitz constant $L_q = \sup_{\|z - z_0\| \leq \bar{r}} \|Dc_q(z)\|_* < \infty$, the local branch-preservation radius is

$$r_{\text{pol}} = \min\left\{\bar{r}, \min_{q: L_q > 0} \frac{m_q}{2L_q}\right\}. \tag{R.A.4}$$

PROPOSITION R.A.2: *Under the hypotheses of Proposition R.A.1, let $S_{bt} = D_z s_{bt}$ and $Q_{bt} = D_z q_{bt}$. The sequence-space path sensitivities satisfy the forward lower-triangular recursion*

$$\begin{aligned} Q_{bt} &= -\mathcal{H}^{-1}_{q,bt}\left(\mathcal{H}_{s,bt} S_{bt} + \mathcal{H}_{z,bt}\right), \\ S_{b,t+1} &= \mathcal{G}_{s,bt} S_{bt} + \mathcal{G}_{q,bt} Q_{bt} + \mathcal{G}_{z,bt}, \end{aligned} \tag{R.A.5}$$

and the simulated moment Jacobian of $\widehat{m}_B(z) = B^{-1} \sum_{b=1}^{B} \Psi(\mathscr{Y}_b(z), z)$ is

$$D_z \widehat{m}_B(z) = \frac{1}{B} \sum_{b=1}^{B} \left[\Psi_{\mathscr{Y},b} D_z \mathscr{Y}_b(z) + \Psi_{z,b}\right]. \tag{R.A.6}$$

Proof. Implicit differentiation of (R.A.3) yields the first equation in (R.A.5), and the chain rule applied to $s_{b,t+1} = \mathcal{G}_{bt}(s_{bt}, q_{bt}, z)$ yields the second, following the sequence-space Jacobian principle of Auclert et al. (2021). Averaging across the B common-random-number histories gives (R.A.6). □

PROPOSITION R.A.3: *Let $\widehat{Q}_B(z) = \widehat{g}_B(z)'W\widehat{g}_B(z)$ denote the finite-sample SMM criterion evaluated by solving the full endogenous model. At a fully evaluated center z_k, any local or fixed-policy directional matrix $\widetilde{J}_k$ used solely to propose candidate steps p_k preserves the exact estimator and asymptotic inference whenever step acceptance ($\widehat{Q}_B(z_k + p_k) < \widehat{Q}_B(z_k)$), stationarity verification, and terminal Jacobians are evaluated using the fully endogenous model.*

Proof. Because $\widehat{Q}_B(z_k + p_k)$, convergence gates, and terminal covariance matrices recompute the 1710 initialization, dated private equilibria, continuation appraisals, and county KKT solutions from scratch, $\widetilde{J}_k$ enters only the step-proposal generator and leaves the scored objective, accepted iterates, and statistical inference invariant (Nocedal and Wright, 2006). □

R.A.3 Parameterization and initialization

As detailed in Online Appendix S.D.1, the parameter vector $\boldsymbol{\psi}$ is mapped to unconstrained optimization coordinates $z \in \mathbb{R}^p$ via logarithmic transforms for positive parameters and hyperbolic-tangent transforms for bounded correlations and shares. The initial 1710 log-productivity profiles $\ln A_{i,1710}(\ell)$ are represented on the five-knot natural cubic spline basis $\{0, 1/4, 1/2, 3/4, 1\}$ and normalized to unit mean $\int_0^1 A_{i,1710}(\ell)\, d\ell = 1$ for $i \in \{A, M\}$. The spatial grid comprises $N = 500$ cells partitioned into $J = 10$ counties, with moments projected onto the 496-cell historical support and twenty-five latitude bands.

R.A.3.1 Spatial grid and conservative finite-volume scheme

The unit interval is partitioned into ordered cells $q \in \{1, \dots, N\}$ with midpoints ℓ_q, widths $\Delta\ell_q > 0$ ($\sum_q \Delta\ell_q = 1$), and county assignment $j(q)$, each county $\mathcal{C}_j$ being a union of whole cells. Cell quantities are evaluated at ℓ_q, and face flows and prices carry half-integer indices. Writing $\kappa_q = \kappa_0 e^{-T_t(\ell_q)}$ and $S_{iq} = Y_{iq} - n_q c_{iq} - e_{iq}\Psi_{iq}/p_{iq} - I_{iq}$ ($I_{Aq} = 0$, $I_{Mq} = I_q$), the conservative discretization of goods conservation (69), spatial no-arbitrage (45), and county service clearing (51) is

$$H_{i,q+1/2} - H_{i,q-1/2} = \Delta\ell_q\big[S_{iq} - \kappa_q|H_{iq}|\big], \tag{R.A.7}$$

$$p_{i,q+1/2} - p_{i,q-1/2} \in \Delta\ell_q(\kappa_q p_{iq} + r_i\tau_{j(q)})\,\mathrm{Sign}(H_{iq}), \tag{R.A.8}$$

$$D_j = \sum_{q\in\mathcal{C}_j}\Delta\ell_q \sum_{i\in\{A,M\}} r_i(Y_{iq} + |H_{iq}|) \le K_j = \sum_{q\in\mathcal{C}_j}\Delta\ell_q Y_{Tq},$$

$$0 \le Y_{iq} \le e_{iq}Z_{iq}G_{iq}L_{iq}^{\alpha_i}, \qquad \tau_j \ge 0, \quad \tau_j(K_j - D_j) = 0, \tag{R.A.9}$$

with closed borders $H_{i,1/2} = H_{i,N+1/2} = 0$, numeraire $p_{A,N+1/2} = 1$, and $\mathrm{Sign}(0) = [-1, 1]$. The cell-centered flow solves $H_{iq} + \frac{1}{2}\Delta\ell_q\kappa_q|H_{iq}| = H_{i,q-1/2} + \frac{1}{2}\Delta\ell_q S_{iq}$, which is strictly increasing in H_{iq}, making the forward flow sweep single-valued whenever

$$\max_q \frac{\Delta\ell_q\kappa_q}{2} < 1, \tag{R.A.10}$$

a condition verified as part of the equilibrium certificate.

Remark R.A.1 (Local branch regularity and spot-market solver). On a fixed complementarity and flow-sign branch with positive prices, wages, capacities, and Stone–Geary slack, $\mathcal{R}_{bt}$ in (R.A.2) is C^1; if $D_u\mathcal{R}_{bt}$ is nonsingular, the implicit function theorem gives a unique local C^1 spot-market branch with sensitivity $\partial_\vartheta u_{bt} = -[D_u\mathcal{R}_{bt}]^{-1}\partial_\vartheta\mathcal{R}_{bt}$ (hypothesis (iv) of Proposition R.A.1; continuum global uniqueness is Proposition 6). Numerically, for each of the $N_\omega = 32$ stratified discovery draws, (R.A.7)–(R.A.9) is solved to relative tolerance 10^{-10} by an active-set Newton–Fischer–Burmeister continuation scheme that alternates forward-backward price-flow sweeps under (R.A.10) with damped Newton updates on the $J = 10$ county toll complementarities, dividend balance, and labor-clearing utility $\bar{u}$.

R.A.3.2 Closed-form quote-to-plan inversion and 1710 initialization

By Propositions 1–2, any statewise quote bank $\{p_{Aq}(\omega), p_{Mq}(\omega), \tau_{j(q)}(\omega), b(\omega)\}$ and candidate 1710 profile $A_{i,1710}(\ell)$ map in closed form into the pre-discovery plan at every cell $q \in \{1, \dots, N\}$:

1. Expected Stone–Geary mobility (18) uniquely determines $\bar{u}$ and local contract wage
$$w_q(\bar{u}) = \frac{\bar{u} + \mathbb{E}_0[(\gamma p_{Aq}(\omega) - b(\omega))/\mathcal{P}_q(\omega)]}{\mathbb{E}_0[1/\mathcal{P}_q(\omega)]}$$
such that total hired labor integrates to $\bar{L}_{1710}$.

2. Given $w_q(\bar{u})$ and expected net unit revenues $\mu_{iq} = \mathbb{E}_0[(p_{iq}(\omega) - r_i\tau_{j(q)}(\omega))_+]$, the bid rents (30)–(32) determine optimal research $\phi^*_{iq} \in [0, 1)$, innovation bill $\Psi_i(\phi^*_{iq}; w_q, \ell_q)$, production labor L^*_{iq}, and sector assignment $e_{iq} \in \{0, 1\}$.

3. Given expected tolls $\bar{\tau}_{j(q)} = \mathbb{E}_0[\tau_{j(q)}(\omega)]$, (50) determines carrier labor L^*_{Tq} and county capacity $K_j = \sum_{q\in\mathcal{C}_j} \Delta\ell_q Y_{Tq}$, while labor clearing fixes resident density $n_q = e_{Aq}L^*_{Aq} + e_{Mq}L^*_{Mq} + L^*_{Tq}$.

Substituting this pre-discovery plan into the spot-market conditions (R.A.11) defines the dated equilibrium operator $\mathcal{T}_{1710}(w, p, \tau) = (w, p, \tau)$ of Definition 1:
$$0 \le Y^0_{iq}(\omega) \le e_{iq}Z^0_{iq}G^0_{iq}(\omega)[L^0_{iq}]^{\alpha_i}, \quad D^0_j(\omega) \le K^0_j, \quad \tau^0_j(\omega) \ge 0, \quad \tau^0_j(\omega)[K^0_j - D^0_j(\omega)] = 0. \tag{R.A.11}$$
Concentrating out the initial productivity spline parameters on the 1710 occupational cross-section via (S.D.2) disciplines the 1710 spatial employment and urban profile. At each date t, the available frontier on the ordered grid is evaluated in $O(N)$ operations per sector via a two-pass dynamic program:
$$F_{iq,t} = \max_{r\in\{1,\dots,N\}} e^{-\delta|\ell_q - \ell_r|}A^+_{ir,t} = \max\{F^{\rightarrow}_{iq}, F^{\leftarrow}_{iq}\}, \tag{R.A.12}$$
where $F^{\rightarrow}_{iq} = \max\{A^+_{iq,t}, e^{-\delta(\ell_q - \ell_{q-1})}F^{\rightarrow}_{i,q-1}\}$ and $F^{\leftarrow}_{iq} = \max\{A^+_{iq,t}, e^{-\delta(\ell_{q+1} - \ell_q)}F^{\leftarrow}_{i,q+1}\}$.

R.A.4 Simulation and optimization sequence

For each specification, the estimator simulates B common-random-number histories over the 172 dates $t \in \{1710, \dots, 1881\}$ ($B_{\text{screen}} = 64$ during multistart screening and $B = 1{,}000$ for final optimization, evaluation, and inference), initializing from a low-discrepancy Halton sequence in the p-dimensional transformed box followed by Gauss–Newton dogleg trust-region refinement. Local stationarity is verified on the minimal normalized positive basis
$$\mathcal{D}_p = \left\{e_1, \dots, e_p, -\frac{1}{\sqrt{p}}\mathbf{1}_p\right\}. \tag{R.A.13}$$

LEMMA R.A.1: $\mathcal{D}_p$ *in* (R.A.13) *positively spans* $\mathbb{R}^p$ *with minimum cardinality* $p+1$*, so for any* $\nabla Q \neq 0$ *at least one* $d \in \mathcal{D}_p$ *satisfies* $\nabla Q' d < 0$*. Supplementing* $\mathcal{D}_p$ *with the* $2p$ *signed-coordinate poll* $\{\pm e_i\}_{i=1}^p$ *guards against finite-step curvature reversals.*

Proof. For any $v \in \mathbb{R}^p$, $t = \max\{0, -\min_i v_i\} \geq 0$ gives $v = \sum_{i=1}^p (v_i + t)e_i + \sqrt{p}\, t(-p^{-1/2}\mathbf{1}_p)$ with nonnegative weights, whereas any p vectors have a pointed nonnegative cone. □

Terminal derivatives use a scaled Helmert regular simplex $D \in \mathbb{R}^{p\times(p+1)}$ with unit columns satisfying

$$D\mathbf{1}_{p+1} = 0, \qquad DD' = \frac{p+1}{p} I_p. \tag{R.A.14}$$

Evaluating the $p+1$ vertices at unit-box radii $h_1 = 0.01$ and $h_2 = 0.005$ with residual matrix G_h and center residual g_0 yields

$$J_h = (G_h - g_0 \mathbf{1}'_{p+1}) D' \left[h \frac{p+1}{p} \right]^{-1}. \tag{R.A.15}$$

Convergence requires that the unit-box gradient satisfy

$$\|\nabla_u Q\|_\infty \leq 5 \times 10^{-4} \max\{1, Q\}, \tag{R.A.16}$$

that no simplex or $2p$ coordinate-poll vertex improves Q by more than $10^{-8}\max\{1, Q\}$, and that the whitened Jacobians $L_Q^{-1} J_{h_1}$ and $L_Q^{-1} J_{h_2}$ ($L_Q L'_Q = W^{-1}$) have full column rank p at tolerance 10^{-10}, singular-value ratio at least 10^{-3}, and relative Frobenius drift at most 0.10. The half-step Jacobian $J = J_{h_2}$ forms $H_{\mathrm{GN}} = 2J'WJ$ and the sandwich covariance (S.D.11).